

# Electroencephalogram silence ratio for early outcome prognosis in severe head trauma

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**Objective:** To introduce the electroencephalogram silence-ratio (ESR) as a variable derived from mathematically processed electroencephalogram for early outcome prognosis in patients with severe head trauma and to comparatively assess sensitivity, specificity and predictive value vs. somatosensory evoked potentials and brainstem auditory evoked potentials.

**Design:** Prospective, interventional study.

**Setting:** Intensive care unit of a university hospital.

**Patients:** A total of 32 adults with severe acute head trauma (Glasgow Coma Scale score  $\leq 8$ ).

**Methods and Main Results:** In all patients, electroencephalographic recording was continuously performed by frontomastoid electrode montage for 24–96 hrs after admission to the ICU. The data were subsequently computed by fast Fourier analysis and the ESR (intervals of suppression as periods  $>240$  msec during which the electroencephalographic voltage did not exceed  $5 \mu V$ ) was displayed and recorded on a computer for further evaluation.

Somatosensory evoked potentials and brainstem auditory evoked potentials were elicited during the first 2 days after admission. Outcome evaluation was performed 6 months after trauma using the Glasgow Outcome Scale and the Rappaport Disability Rating Scale. After careful artifact exclusion, the ESR depicted the highest sensitivity, specificity, and positive predictive value compared with evoked potentials. Even a highly significant correlation between outcome and ESR was found ( $p < .0001$ ).

**Conclusion:** The ESR is a valuable variable showing a high reliability with respect to outcome prediction in severe head trauma with a higher predictive value than short latency somatosensory evoked potentials. Evidence exists that the ESR provides at least partial information regarding adequate cerebral oxygen delivery. (Crit Care Med 2000; 28:3522–3529)

**KEY WORDS:** head trauma; outcome; processed electroencephalogram; neuromonitoring; electroencephalogram-silence ratio; somatosensory evoked potentials; brainstem evoked potentials

Evidence has been found that electroencephalography can provide important prognostic information in patients with severe head trauma as stressed by Jordan (1). However, the pathologic alteration in the electroencephalogram induced by traumatic brain injury and rated for outcome prediction varies considerably. Burst suppression pattern is well known to correspond with poor outcome, although it is most commonly a consequence of hypoxic or metabolic brain damage that might have occurred after trauma (2). Gütling et al. (3) found a high

correlation between electroencephalographic (EEG) reactivity to external stimuli and outcome after head injury whereas Evans and Bartlett (4) emphasized that recognition of sleep-related activity was followed by an adequate neurologic state. Rae-Grant et al. (5) developed a scale for outcome prediction rating various EEG abnormalities from previous investigations. However, all data were obtained from raw electroencephalograms, which are frequently difficult to interpret, especially for inexperienced observers. It would be useful if a metric variable derived from raw electroencephalograms could provide accurate information for early outcome prognosis. In several investigations, low amplitude EEG events have been found to correlate with poor outcome after brain injury (6–8). Rae-Grant et al. (6) described episodic low-amplitude events lasting 0.5–4 secs with EEG total power  $<5 \mu V$  showing different background activities. Synek and Synek (7) detected the same phenomenon combined with theta activity only. Brenner et al. (8) reported intermittent EEG inactivity lasting for seconds during

$\delta$ EEG. All authors stressed that these patterns can be clearly distinguished from burst suppression. The question arises whether this electrophysiological occurrence may be quantifiable and subsequently able to provide valuable information regarding early outcome prediction in patients with head trauma. Rampil et al. (9) introduced a variable called “burst suppression ratio” for quantifying the burst suppression phenomena induced by narcotic drugs. The algorithm described intervals of suppression as periods  $>240$  msec during which the EEG voltage did not exceed  $5 \mu V$ , then calculated as percentage of a given time period. This variable might be suitable for assessing quantitatively low-amplitude events in the electroencephalogram.

The present study was performed to investigate whether the “burst suppression ratio” might be a useful variable for early outcome prognosis in patients with severe head trauma. To avoid misunderstandings with respect to classic burst suppression pattern, we renamed the term “burst suppression ratio” as “electroencephalogram silence ratio” (ESR),

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which might be a more accurate term for the introduced variable with respect to its application. Sensitivity, specificity auditory evoked potentials (BAEP), and positive predictive value of the ESR were compared with that of brainstem auditory evoked potentials (BAEP) and short-latency somatosensory evoked potentials (SSEP).

## PATIENTS AND METHODS

**Patient Population.** The prospective study was undertaken at a university intensive care unit from July 1996 to June 1998. The study group consisted of 45 patients (35 males, 10 females) with severe head injury after closed head trauma. The mean age was  $35.2 \pm 17.9$  yrs (range, 18–75 yrs). All patients presented with a Glasgow Coma Scale (GCS) score  $\leq 8$  after resuscitation or deteriorated to this level within 6 hrs after trauma (median, 5). Informed consent for participation in this study was obtained from next of kin.

Diagnostic cranial computed tomographic (CT) scans were obtained in all patients at hospital admission, followed by immediate surgical intervention in 24 patients. To additionally describe the brain injury pattern, the Marshall CT scan classification (10) was used.

In addition, in eight healthy awake volunteers (mean age,  $26.5 \pm 8.5$  yrs [mean  $\pm$  sd]) and in nine patients without brain injury, metabolic disorders, or sepsis that might influence electrocortical activity substantially who were sedated by midazolam/fentanyl for  $\geq 24$  hrs ( $38.4 \pm 12.8$  yrs), the ESR was measured.

**Clinical Management.** Multimodality physiologic monitoring involved electrocardiography, arterial blood pressure by peripheral arterial cannulation, arterial oxyhemoglobin saturation by transcutaneous pulse oximetry, body temperature, and mean intracranial pressure (ICP). Mechanical ventilation was controlled by repeated blood gas analysis. ICP monitoring was performed via epidural or intraventricular catheterization. Central venous catheters were placed in all patients to guide normovolemic fluid management.

After ICU admission, sedation was performed by continuous application of midazolam ( $1.8\text{--}10.4 \mu\text{g/kg/min}$  iv) in combination with fentanyl ( $3.7\text{--}11.1 \text{ ng/kg/min}$  iv) for analgesia. Cerebral perfusion pressure (CPP) was held at  $\geq 70$  mm Hg. In 19 patients, a continuous intravenous application of norepinephrine was therefore necessary.

If ICP values exceeded 20 mm Hg for  $>10$  mins, the treatment of the intracranial hypertension was supplemented in the following manner in addition to the previously induced sedation: head elevation to between  $20^\circ$  and  $30^\circ$  from horizontal while maintaining an adequate CPP, administration of osmotic diuretics (mannitol), and mild hyperventilation ( $\text{Paco}_2$ ,  $37.5 \pm 1.5$  mm Hg).

In 14 patients, a further elevation of ICP to values  $>30$  mm Hg was followed by decompressive craniectomy. If this therapy failed, barbiturates were applied intravenously. The algorithm fits the procedure of ICP therapy as suggested by Maas et al. (11).

**Electroencephalographic Measures.** The EEG recording was performed using a computerized two-channel EEG monitor (pEEG-Monitor, Dräger, Lübeck, Germany) that displays both the raw EEG signal from the frontal brain and the computed data. Electrodes were placed on the scalp with bipolar frontomastoid montage (Fp1-A1 and Fp2-A2; international 10–20-system of electrode placement) with central forehead ground. Conventional Ag-AgCl electrocardiographic electrodes were taken for EEG recording. The EEG data were obtained at a sampling rate of 128 Hz and subsequently analyzed continuously during 2-sec data segments. Analysis was performed by fast Fourier transformation and power spectral analysis as described elsewhere (12). The skin was processed to reach an impedance of  $<4 \text{ k}\Omega$  at each electrode. Every 5 mins the impedance was controlled automatically by the system and impedance shifts were adjusted by a replacement of electrodes. Amplifier bandwidths were 1.5–30 Hz, and the amplifier gain was set at 4300. The forehead electromyogram was recorded simultaneously for artifact rejection.

The monitoring was started  $\leq 12$  hrs after trauma and was continued for  $\geq 24$  hrs up to 96 hrs. The mean duration of monitoring was  $58 \pm 19$  hrs. Monitoring was discontinued earlier than 96 hrs when sedation was stopped and EMG effected the ESR artificially or barbiturates or propofol were injected to decrease high intracranial pressure levels.

If any influence of muscular activity or eye movement on the electroencephalogram could not be excluded, a short acting neuromuscular blocking agent was applied intravenously.

**EEG-Silence Ratio.** The EEG-silence ratio was defined in accordance to the definition of Rampil et al. (9) as intervals of suppression as periods  $>240$  msec during which time the EEG voltage does not exceed  $5 \mu\text{V}$ . EEG silence was determined by the system if  $\geq 32$  successive samples of 128 sampling points/sec were  $<5 \mu\text{V}$  and subsequently averaged for a time of 60 secs. If substantial interhemispheric differences occurred, the lower value of both hemispheres was taken for further outcome evaluation.

**Monitoring of Short-Latency Somatosensory Evoked Potentials and Brainstem Auditory Evoked Potentials.** SSEP and brainstem auditory evoked potentials (BAEP) were obtained at the bedside with the Neuropack II instrument (Nihon-Kohden, Tokyo, Japan) at least during the first 2 days after ICU admission.

The SSEP were elicited by electrical stimulation of the median nerves at 5 Hz bilaterally recording at second cervical level (N13

peak) and parietal scalp (N20 peak). A total of 200 responses were averaged per trial. N20 latency and central conduction time were estimated and taken for further evaluation. The results were scored in a 3-grade-scale according to Judson et al. (13) as follows: 1, normal symmetrical; 2, increased central conduction time on one or both sides; 3, absent cortical responses over one or both hemispheres. BAEP were elicited by 10 Hz alternating click stimulation of each ear (100-dB intensity) recorded at the vertex scalp with ipsilateral mastoid reference. A total of 1,200 responses were averaged per trial. The results were scored according to Cant et al. (14) as follows: 1, normal bilaterally; 2, abnormal latencies, wave V present; 3, wave V absent on one or both sides.

**Outcome Evaluation.** Outcome was assessed 6 months after discharge from ICU by means of the Glasgow Outcome Scale (GOS) (15) and the Rappaport Disability Rating Scale (RDRS) (Table 1), which was introduced to provide a more specific outcome evaluation with regard to neuropsychological behavior (16).

The evaluations were obtained either by immediate contact with the patients or by telephone interview with the patients or their guardians or caretakers. To standardize the estimates of the patient status, the interviews

Table 1. Rappaport Disability Rating Scale (15)

|                                     |   |
|-------------------------------------|---|
| Eye opening                         |   |
| Spontaneous                         | 0 |
| To speech                           | 1 |
| To pain                             | 2 |
| None                                | 3 |
| Best motor response                 |   |
| Obedient                            | 0 |
| Localizing                          | 1 |
| Withdrawing                         | 2 |
| Flexing                             | 3 |
| Extending                           | 4 |
| None                                | 5 |
| Self care items                     |   |
| Complete                            | 0 |
| Partial                             | 1 |
| Minimal                             | 2 |
| None                                | 3 |
| Employability                       |   |
| Not restricted                      | 0 |
| Selective workshops, competitive    | 1 |
| Sheltered workshops, noncompetitive | 2 |
| Nonemployable                       | 3 |
| Level of functioning                |   |
| Completely independent              | 0 |
| Independent in special environment  | 1 |
| Mildly independent                  | 2 |
| Moderately dependent                | 3 |
| Markedly dependent                  | 4 |
| Totally dependent                   | 5 |
| Best verbal response                |   |
| Oriented                            | 0 |
| Confused                            | 1 |
| Inappropriate                       | 2 |
| Incomprehensive                     | 3 |
| None                                | 4 |

were structured with yes/no answers to all categories used in the RDRS.

**Exclusion Criteria.** Patients were excluded from the study if the raw electroencephalogram showed electrocerebral inactivity or a burst-suppression-pattern initially or during the first 2 days of treatment (n = 6). In patients presenting with these patterns, a poor outcome can be expected without exception. New information regarding outcome evaluation could not be expected by use of the ESR. Although no patients screened for the study showed metabolic disorders or sepsis, which might substantially effect the ESR, such occurrences would have necessitated exclusion.

The immediate use of barbiturates, propofol, or  $\gamma$ -hydroxybutyrate to treat intracranial hypertension led to the exclusion of seven patients from the study. The application of these drugs, even in clinically relevant dosages, is able to increase the ESR in a substantial manner, in contrast to benzodiazepines.

**Statistical Analysis.** For statistical analysis and assessment of significance, we calculated Pearson's correlation coefficients, linear, and

multivariate forward, stepwise logistic regression analysis. In addition predictive accuracy was computed to clarify the association between outcome and GCS score, somatosensory evoked potentials, ESR, and age. By means of cross tables we additionally computed sensitivity, specificity, and positive predictive values for ESR, SSEP and BAEP. Student's *t*-test

was taken for analysis of variance. A  $p < .05$  was considered statistically significant.

## RESULTS

**Outcome Measurements.** As demonstrated in Figure 1, 22 patients (69%) showed a favorable outcome (GOS, 4–5;

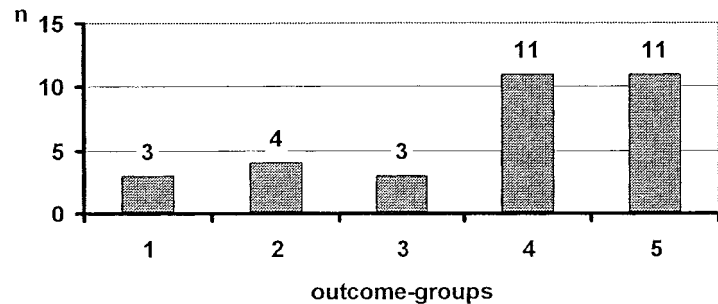


Figure 1. Outcome of 32 comatose patients suffering from severe brain trauma according to Glasgow Outcome Scale.

Table 2. Individual patient data describing computed tomographic scan findings, necessity of surgical intervention, the computed score of the Marshall Classification (10), and the Glasgow Coma Scale Score at admission to hospital; data are subdivided into patients with good (n = 22) and poor outcome (n = 10)

| Patient No.    | Epidural Hematoma | Subdural Hematoma | Contusion | SAH Traumatic | Cisterns   | Midline-shift >5mm | Surgical Intervention | Marshall Category | GCS Score |
|----------------|-------------------|-------------------|-----------|---------------|------------|--------------------|-----------------------|-------------------|-----------|
| <b>GOS 4–5</b> |                   |                   |           |               |            |                    |                       |                   |           |
| 1              | Y                 | N                 | N         | N             | Present    | N                  | n                     | 2                 | 6         |
| 2              | N                 | N                 | Y         | N             | Present    | N                  | n                     | 2                 | 8         |
| 3              | N                 | N                 | Y         | N             | Absent     | N                  | n                     | 3                 | 5         |
| 4              | Y                 | Y                 | N         | Y             | Compressed | N                  | n                     | 3                 | 7         |
| 5              | N                 | N                 | N         | Y             | Compressed | N                  | n                     | 3                 | 8         |
| 6              | N                 | Y                 | Y         | N             | Compressed | N                  | n                     | 3                 | 6         |
| 7              | Y                 | Y                 | Y         | N             | Absent     | Y                  | y                     | 4                 | 7         |
| 8              | N                 | Y                 | Y         | N             | Absent     | Y                  | y                     | 4                 | 3         |
| 9              | N                 | Y                 | Y         | N             | Absent     | Y                  | y                     | 4                 | 8         |
| 10             | Y                 | N                 | N         | N             | Compressed | Y                  | y                     | 4                 | 3         |
| 11             | N                 | N                 | Y         | N             | Compressed | N                  | y                     | 4                 | 8         |
| 12             | Y                 | N                 | N         | N             | Compressed | Y                  | y                     | 4                 | 7         |
| 13             | N                 | Y                 | N         | N             | Compressed | Y                  | y                     | 4                 | 5         |
| 14             | N                 | Y                 | N         | N             | Compressed | N                  | y                     | 4                 | 7         |
| 15             | N                 | Y                 | Y         | Y             | Absent     | Y                  | y                     | 4                 | 5         |
| 16             | Y                 | N                 | N         | N             | Compressed | Y                  | y                     | 4                 | 8         |
| 17             | N                 | Y                 | Y         | N             | Absent     | Y                  | y                     | 4                 | 3         |
| 18             | Y                 | N                 | N         | N             | Compressed | Y                  | y                     | 4                 | 8         |
| 19             | Y                 | N                 | N         | Y             | Compressed | Y                  | y                     | 4                 | 3         |
| 20             | N                 | N                 | Y         | Y             | Absent     | Y                  | n                     | 4                 | 5         |
| 21             | N                 | Y                 | Y         | N             | Absent     | Y                  | y                     | 4                 | 6         |
| 22             | N                 | Y                 | Y         | N             | Absent     | N                  | y                     | 4                 | 3         |
| <b>GOS 1–3</b> |                   |                   |           |               |            |                    |                       |                   |           |
| 1              | N                 | N                 | Y         | N             | Compressed | N                  | n                     | 3                 | 3         |
| 2              | N                 | N                 | Y         | N             | Compressed | N                  | y                     | 4                 | 6         |
| 3              | N                 | N                 | Y         | Y             | Compressed | Y                  | y                     | 4                 | 5         |
| 4              | N                 | Y                 | Y         | N             | Compressed | Y                  | y                     | 4                 | 4         |
| 5              | N                 | Y                 | Y         | N             | Absent     | Y                  | y                     | 4                 | 3         |
| 6              | Y                 | N                 | Y         | Y             | Absent     | Y                  | y                     | 4                 | 3         |
| 7              | Y                 | N                 | Y         | N             | Compressed | Y                  | y                     | 4                 | 3         |
| 8              | N                 | Y                 | Y         | N             | Compressed | Y                  | y                     | 4                 | 3         |
| 9              | N                 | N                 | Y         | Y             | Compressed | Y                  | y                     | 4                 | 3         |
| 10             | N                 | Y                 | Y         | Y             | Absent     | Y                  | y                     | 4                 | 3         |

SAH, subarachnoid hemorrhage; GOS, Glasgow Outcome Scale score; GCS, Glasgow Coma Scale; Y, present; N, Not present, y, performed; n, not performed.

“Contusion” does not differentiate between patients with or without contusional bleeding.

RDRS, <15) 6 months after injury in contrast to 10 patients (31%) who were severely disabled, vegetative, or died in the course of the disease (GOS, 1–3; RDRS >15). The median of the initial GCS score in all patients with good outcome was 6 whereas in the group showing poor outcome, the median GCS score was 4. No statistically significant difference was found with regard to the mean age of both groups ( $30.8 \pm 14.6$  vs.  $35.9 \pm 19.1$  yrs). Table 2 presents the individual patient data with the CT scan findings obtained immediately after admission to hospital.

**Evoked Potentials.** In Figure 2, the distributions of the SSEP and BAEP grades following the scores introduced by Judson et al. (13) and Cant et al. (14) are presented separately for both groups with good and poor neurologic outcome. According to the supposition of Cant et al. (14), SSEP showing grades 2 or 3 were characterized as abnormal, thus indicat-

ing a poor neurologic outcome. Using this classification, sensitivity for SSEP was calculated at 0.8, specificity at 0.82, and the positive predictive value at 0.66. Concerning BAEP, sensitivity was 0.47, the specificity was 0.64, and the positive predictive value was found at 0.47.

**EEG-Silence-Ratio.** Figure 3 presents the RDRS and GOS plotted as a function of the ESR including linear regression lines. We observed a highly significant relationship ( $p < .001$ ) between RDRS vs. ESR ( $r^2 = 0.66$ ) and GOS vs. ESR ( $r^2 = 0.67$ ). The ESR in patients with GOS 4–5 was  $10.0\% \pm 10.8\%$ , whereas in patients with poor outcome ESR was computed at  $51.7\% \pm 20.0\%$  (significant for  $p < .0001$ ). As is visible in both figures, an ESR increase implies a higher probability of a poor outcome. Comparing outcome measures with ESR, a critical range of approximately 20% to 25% was concluded. Merely 1 of 22 patients showing an adequate outcome presented with ESR

values >25%. In this patient, the ESR was observed at 50.2% despite an adequate outcome 6 months after injury (GOS score, 5). In this patient, substantial bifrontal cerebral contusions caused an attenuation of frontal EEG activity, thus inducing an inadequate high ESR.

In patients considered for the study displaying poor outcome ESR values >20% were observed without exception. However, one patient kept out of the study as a result of exclusion criteria showed an ESR <20% although exhibiting a vegetative state 6 months after injury. In this patient, the nuclear magnetic resonance disclosed an isolated small pontine infarction not visible in the CT scan.

The RDRS has been developed to improve the assessment of neuropsychological disorders after head trauma compared with GOS (16). Plotting the RDRS value as a function of GOS score (Fig. 4), we found that patients showing a poor outcome with GOS scores of 1–3 always showed a RDRS value >14. On this account, a RDRS value <15 was rated as good neurologic outcome and taken for further calculation of sensitivity, specificity, and positive predictive value. Cross-table analysis (ESR <20% and ESR >20% vs. GOS scores 4–5 vs. GOS scores 1–3; ESR <20% and ESR >20% vs. RDRS <15 vs. RDRS >15) revealed a high sensitivity (0.91) and specificity (0.91) with a positive predictive value of 0.83.

Stepwise forward, multiple logistic regression showed that only the ESR ( $p < .01$  with an accuracy of 90.6%) and SSEP ( $p < .01$ , accuracy of 84.4%), but neither GCS score ( $p > .1$ , accuracy of 75%) nor age ( $p > .1$ , accuracy of 68%) is an independent predictor of outcome. The connected predictive information of ESR and SSEP, ESR and GCS score, or ESR and age only showed a small increase in predictive accuracy using ESR and SSEP (93.2%) but no improvement for other combinations.

By a subgroup analysis of all patients who underwent decompressive craniectomy, we even found a high correlation between ESR values and GOS ( $r^2 = 0.53$ ;  $p < .001$ ) or RDRS ( $r^2 = 0.61$ ;  $p < .001$ ). In two of six patients with an initially high ESR after decompressive craniectomy, a substantial decrease was observed immediately after surgery.

The ESR in eight healthy awake volunteers was constantly zero. In nine patients sedated by midazolam/fentanyl, the

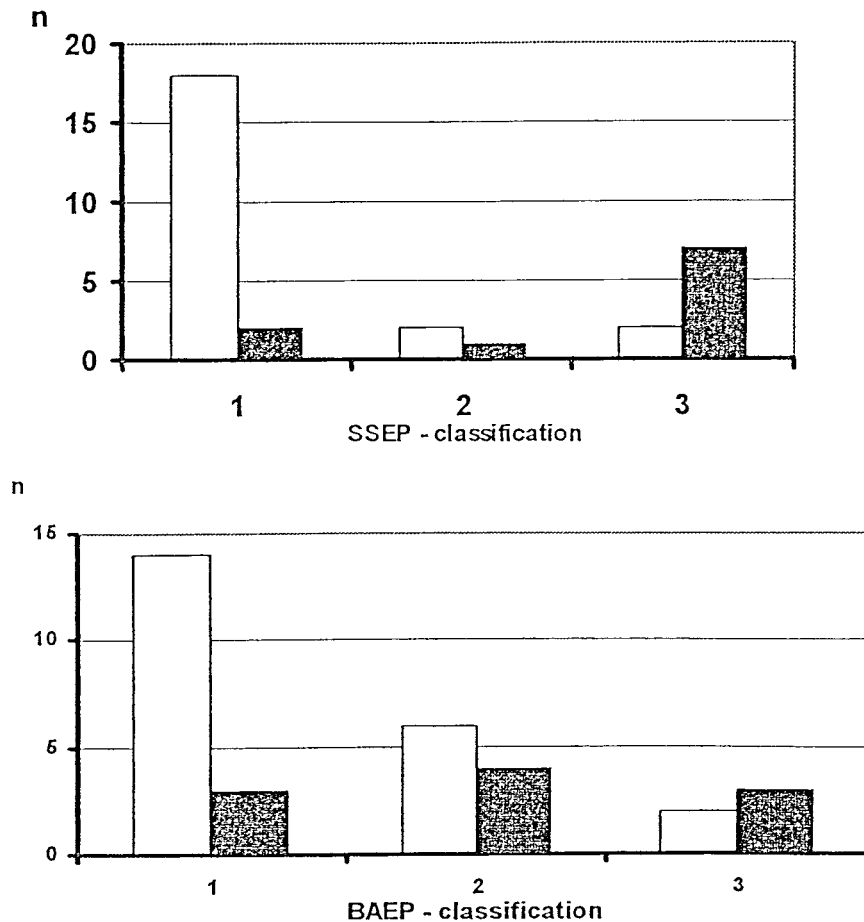


Figure 2. Top, relationship between short-latency somatosensory evoked potentials (SSEP) and outcome in 32 patients scored according to the grading of Judson et al. (13) (white bars, GOS scores of 4–5; black bars, GOS scores of 1–3). Bottom, relationship between brainstem auditory evoked potentials (BAEP) and outcome in 32 patients scored according to the grading of Cant et al. (14) (white bars, GOS scores of 4–5; black bars, GOS scores of 1–3).

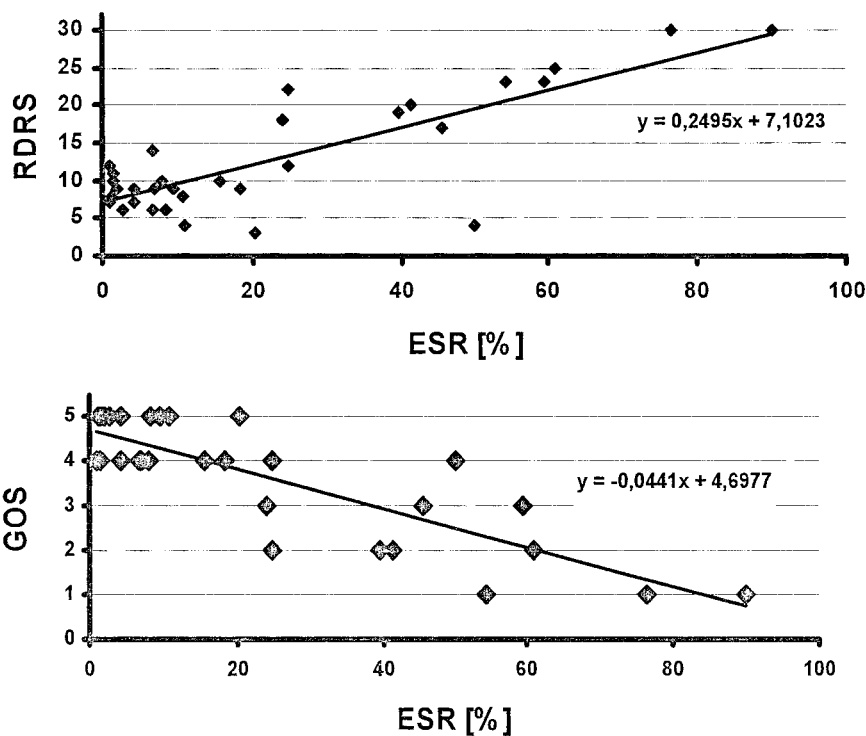


Figure 3. Rappaport Disability Rating Scale (RDRS) (top) and Glasgow Outcome Scale (GOS) (bottom) scores plotted as a function of the electroencephalogram-silence ratio (ESR). In both graphs, regression lines including mathematical formulas are presented. A highly significant correlation was found for both outcome variables ( $p < .0001$ ) compared with ESR. The critical range of the ESR begins at 20% to 25%.

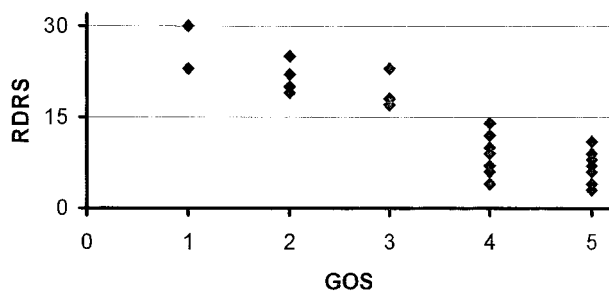


Figure 4. Rappaport Disability Rating Scale (RDRS) compared with Glasgow Outcome Scale (GOS) values. Patients with GOS scores  $\geq 4$  exclusively showed a RDRS value  $< 15$ .

ESR was  $< 5\%$  although the serum concentration of midazolam ranged from 450 ng/mL to 890 ng/mL.

## DISCUSSION

According to the results of the present study we are able to introduce the EEG-silence ratio as a highly feasible variable for outcome prediction in severe head trauma. Because the ESR is a metric variable derived from the raw electroencephalogram, it can be interpreted even without being familiar with special EEG analysis. This may induce a wider application of the mathematically processed electroencephalogram in clinical routine.

To achieve a high reproducibility of the approach, some important methodologic principles have to be stressed. The amplitude of the EEG signal can be artificially affected by several methodologic alterations. It is well known that an increase in the electrode distance results in an augmentation of the EEG amplitude (17). Correspondingly a closer electrode montage leads to an artificial increase of the ESR as it could be observed frequently. The frontomastoid montage chosen for ESR estimation could not be changed without affecting the ESR. Even a change of the electrode montage to the occipital or temporal cortex is able to

According to the results of the present study we are able to introduce the electroencephalographic silence ratio as a highly feasible variable for outcome prediction in severe head trauma.

modulate the EEG amplitude (18) and hence the ESR. Thatcher et al. (19), however, who examined the suitability of the electroencephalogram for outcome prediction by using different electrode montages observed that frontal scalp leads, which were chosen for the present study, were the most significant predictors of outcome.

Another important component is the choice of the sedatives and analgesic drugs. Benzodiazepines in dosages commonly used for adequate sedation do not result in an attenuation of the EEG amplitude but were described to increase EEG activity in the  $\beta$ -frequency range, especially in frontal brain areas (20). For opioids, an increase in the EEG amplitude accompanied by a slowing has been observed in several studies (21, 22). These drugs do not flatten the EEG amplitude although Herkes et al. (23) described the development of a burst suppression pattern in septic patients while applying midazolam. However, this phenomenon could also be ascribed to hypoxic brain damage that might have occurred in these patients in the course of the disease. In eight healthy young persons and in nine patients without cerebral injury sedated by midazolam and fentanyl, an ESR increase was not found in the present study. In contrast, immediately after the intravenous application of barbiturates, propofol,  $\gamma$ -hydroxybutyrate, or etomidate, an artificial increase of the ESR occurred systematically in a dose-dependent manner. Consequently, the accurate interpretation of the electroencephalogram after the use of these drugs is impossible until it is completely metabolized or eliminated from the body. This might imply a limitation of the method. However, the combination of

benzodiazepines with opioids is the most commonly used sedative strategy in most intensive care units during the initial phase of head trauma. This fact confines this methodologic insufficiency to patients requiring extended pharmacologic therapy for the treatment of a strongly increased intracranial pressure. In our study, 16% of all patients initially considered for the study had to be excluded because of prolonged barbiturate use started immediately after admission to ICU that was caused by an initially extensive ICP. This value is comparable to currently available data. In several studies, 10% to 12% of head-injured patients manifested intracranial pressure levels suggesting the additional use of barbiturates for ICP lowering (24–27). Taking into account that the use of barbiturates or similar drugs for ICP therapy is presently not a common practice (28), ESR monitoring is impossible because of barbiturate application only in a small subset of patients.

Even the exclusion of patients presenting with sepsis or metabolic disturbances only concerns a very small group of brain traumatized patients. As reported by Piek et al. (29) septicemia and renal, hepatic, or gastrointestinal abnormalities are either late complications of head trauma patients not effecting early prognosis by use of the ESR or only a very small subgroup with inconsiderable influence on methodologic suitability. We did not exclude one patient as a result of metabolic disorders.

The exclusion of patients with burst suppression electroencephalogram or EEG inactivity was done to make higher demands on the presented method. If a native electroencephalogram is available, these patterns can be detected and interpreted without using the ESR. If not, however, the ESR is able to provide reliable information. All ESR values were measured distinctly >25% while showing a poor outcome. Including the six patients initially presenting with EEG inactivity or burst suppression electroencephalogram, a higher correlation between ESR values and outcome would be computed.

There is evidence that the increasing reduction of the EEG amplitude even as brief intermittent events characterizes a more appreciable injury of the cerebral integrity. In the EEG abnormality grades developed by Synek (30), the EEG amplitude is more attenuated in patients with a poor outcome. Even low amplitude

events have been described by several authors to imply a poor prognosis (6–8). However, in all studies it has been a rare event. Rae-Grant et al. (6) detected “episodic-low-amplitude-events” in 20 of 2,730 electroencephalograms analyzed

over 2 yrs. In our group, >30% of all patients displayed very low amplitude EEG events. The discrepancy might be explained by scrutinizing the raw electroencephalogram of a 32-yr-old patient suffering from severe head injury in com-

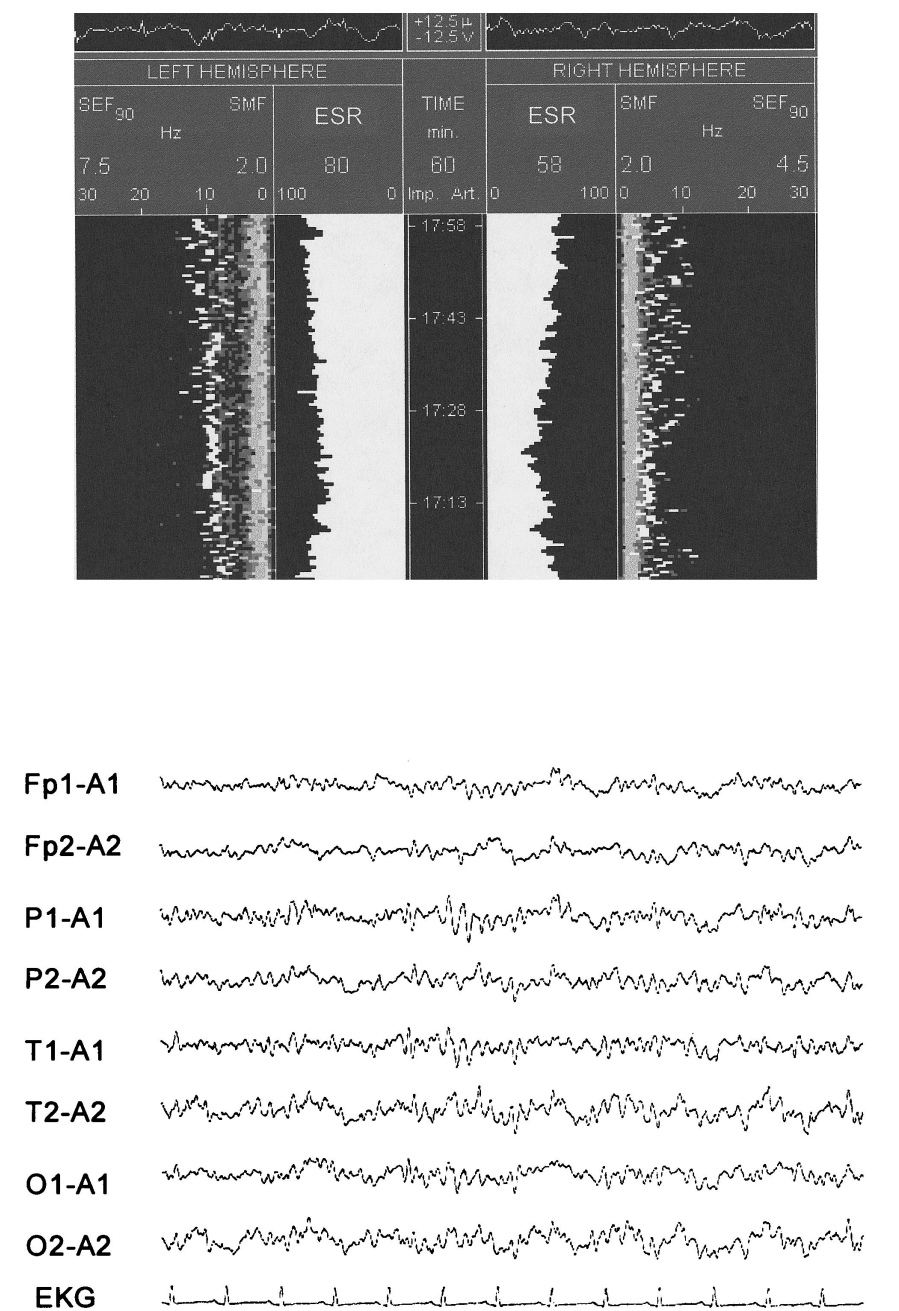


Figure 5. Top, display of the computerized monitoring systems for electroencephalogram-silence ratio evaluation. On top, the raw electroencephalogram line of one epoch is visible to realize possible artifacts (electrocardiogram, electromyogram). Below, a 60-min recording of the spectral edge frequency (SEF<sub>90</sub>) and the electroencephalogram-silence ratio are displayed separately for both hemispheres. The presented patient suffered from a severe brain trauma (GCS score, 5) with a massive subdural hematoma on the right side. The high electroencephalogram-silence ratio value indicated a poor outcome, which was confirmed 6 months later while still exhibiting a vegetative state (display modified by courtesy of Draeger AG, Lübeck, Germany). Bottom, raw electroencephalogram of the same patient obtained during the same time as the top figure. No clear low-amplitude event can be detected by eye inspection.

parison with the EEG signal of the mathematically processed electroencephalogram (Fig. 5). According to the definition of Rae-Grant et al. (6), no definite episodic low-amplitude events can be observed. In contrast, the simultaneously obtained processed electroencephalogram displayed an ESR of nearly 50% and a poor outcome of this patient was found 6 months after trauma (GOS score, 2). Because the processed electroencephalogram is able to identify low amplitude events lasting 0.25 secs, this method is certainly superior to conventional visual inspection of a raw electroencephalogram. This might be the explanation for substantial differences of the incidence of the described pathologic aspect of the present study compared with earlier investigations.

The question arises which kind of pathophysiological background might explain the described phenomenon. An increased ESR representing a profound disruption of normal electrophysiological activity might indicate either structural or functional involvement of cortical or subcortical structures. Morphologic lesions of the brain like diffuse axonal injury, especially in the reticular formation, are able to affect both electrocortical activity and conscious behavior or arousal (31, 32). This injury pattern has even been a repeated, although not consistent, finding of magnetic resonance imaging diagnosis in our patients showing high ESR values (6 of 10 patients). Even widespread cortical contusions in frontal brain areas are able to influence the ESR substantially. However, this damage is even able to cause incorrectly high values referred to prognosis, as observed in one patient of our study. In this patient, the ESR was measured at nearly 50% despite an adequate outcome (GOS score, 5). Consequently, it must be outlined that the introduced variable, like any other electrophysiological variable, must be interpreted in the entirety of all other diagnostic findings to reach a maximum of reliability with regard to its approach.

An EEG amplitude lowering could also be the result of a functional alteration of electrophysiological activity. It has been a frequent finding that cerebral ischemia occurring early after head trauma and probably induced by cerebral arterial spasm is a significant complication and can lead to impaired cerebral blood flow, especially in association with traumatic subarachnoidal hemorrhage (33–35). It is even a well known fact that electrophys-

iological activity is linked to cerebral metabolism and oxygen delivery as summarized by Jordan (1). It is necessary, however, to distinguish exactly between different extents of the impaired oxygen delivery. As postulated by Speckmann and Caspers (36) and confirmed in a recent study of Goel et al. (37), during moderate hypoxia medium- and high-frequency components of the electroencephalogram increased whereas low-frequency components declined, accompanied by a reduction of the total power. This response was explained by an altered DC potential in the cortical areas. A further deterioration of oxygen delivery leads to EEG slowing and subsequently to EEG silence. It can be hypothesized that the ESR reflecting at least partially the EEG amplitude might already be affected even by moderate hypoxia. It has been noticed frequently in our patients who an acute reduction of the CPP resulted in an ESR increase before a decline of EEG frequency (own unpublished observations). Proceeding from the assumption that an elevated ESR is able to reflect cerebral hypoxia, an improvement of oxygen delivery should have a significant effect on this variable. In fact, we could reduce the ESR considerably while increasing cerebral perfusion pressure or adjusting intravascular volume. In an accompanying (unpublished) study, we were able to decrease the ESR in 36% of all patients with initially pathologic values. All patients, even those with a primarily high ESR which could be lowered to <20%, showed GOS scores of 4–5 6 months after injury. While observing the ESR reactivity after clamping of one internal carotid artery during carotid endarterectomy, a strong ESR increase in the ipsilateral brain hemisphere was observed as a repeated finding. Because even the blood flow velocity in the ipsilateral middle cerebral artery estimated by transcranial Doppler sonography declined markedly, we concluded that this reaction was a consequence of a pathologically relevant reduction of oxygen delivery. Declamping normalized the ESR and blood flow velocity quickly.

After decompressive craniectomy in two of six patients showing a pathologically high ESR value, a considerable decline of ESR was observed immediately after surgery, which can be ascribed to a substantial increase of the CPP thus resulting in an improvement of cerebral oxygen delivery. It cannot be completely excluded that even a change of electrical

properties (such as impedance or capacitance) or the development of a breach rhythm might affect the ESR. Regarding the highly predictive information of the ESR in these patients, however, it is reasonable to assume that low amplitude episodes detected by the ESR specifically reflected brief episodes of very low activity of cortical neurons.

Every new variable introduced in clinical routine should provide improvements compared with established methods. Accordingly, the ESR as a prognostic variable must be matched with SSEP, which has been proven to be the most useful variable for outcome prediction in sedated brain trauma patients. The prognostic accuracy of SSEP in the present study was analogous to those of previous studies (13, 14, 38, 39) as well as the superiority compared with BAEP (13). Comparing this issue with the ESR, the new variable was found to be at least as feasible as SSEP. In one patient, it even contradicted the findings of SSEP. Despite initially lost cortical responses in both hemispheres, a very low ESR questioned this finding. In the course of the disease, the prognostic assessment of the ESR could be confirmed. The patient regained consciousness and 6 months after injury exhibited a GOS score of 4. SSEP reappeared 3 wks after trauma. However, as mentioned above in another case, the SSEP were superior to ESR with regard to outcome prediction. In one patient showing isolated infratentorial infarction, cortical responses were absent in both hemispheres despite a low ESR value. This limitation of the introduced approach was confirmed in two patients not included in the present study suffering from thrombosis of the basilar artery. During the first days even in these cases, the ESR showed values that have to be interpreted as not critical (unpublished observations). However, the limitation of predictive value of the ESR is limited to patients with isolated infratentorial lesions. On the other hand, the ESR can possibly be taken as a diagnostic tool; this must be confirmed in a future study.

Another advantage is the possibility of a continuous noninvasive monitoring of brain integrity. As described above, the ESR is not always a steady value but was sometimes observed to vary during the monitoring period. This was possibly induced by different levels of oxygen delivery, as discussed above. Further investigations should be performed to evaluate the suitability of this device to provide an

additional tool for controlling adequate cerebral oxygenation.

As an additional convenience, the ESR is not economically burdensome. Many computerized EEG systems already integrated the burst suppression ratio, which is a synonym for the ESR. Furthermore, in the device used for the present study it is even possible to take conventional electrocardiographic electrodes for EEG monitoring, which allows a continuous evaluation of cortical activity for days without any risk of local infection. A further attractive feature is the simplicity of interpretation, because the ESR as an intelligible metric variable can be valued easily and quickly even for the inexperienced observer with regard to electroencephalography.

In conclusion, the presented method is a suitable, highly reliable, and practical alternative for outcome prediction in severe head trauma. Furthermore, several advantages have been described that demonstrate that this approach provides important information in comatose patients, thus making it an effective noninvasive monitoring variable for neurophysiological monitoring on intensive care units. Our data suggest that this variable should be included in multimodality neuromonitoring in patients with head trauma.

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